Assignment #4

Motion Detector

Submit By: 11:59pm October 6, 2024
revised September 23, 2024

1 Overview

For this assignment your goal is to build a motion detector! It should detect motion from a webcam or video stream and then visualize it by drawing a shape over the video stream.

2 Instructions

1. Use some sort of preprocessing on your image before feeding it into your motion detector.
2. Use one of the foreground detection methods mentioned in class.
3. Use your motion detector to draw a box/circle around the motion. **You may not use contours for this.** Too easy.
4. Record a video of your motion detector in action. You can choose how to do this, to use OpenCV see the tutorials here, using `cv.VideoWriter(...)`.
5. Write-up:
   - For your write-up, explain what motion detection algorithm(s) you used with a brief summary of how it works.
   - Also include any assumptions you are making. These would include assumptions that simplify the problem you are trying to solve, such as environment constraints.

3 Requirements

Make sure your zipped submission contains:

1. Source code `.py` file *with your name at the top.*

2. The video recording of your detector in action.

3. Your write-up as a PDF.

    - ***Note:*** *If you are looking for an easy way to convert your write-up from something like Markdown to PDF, you might be interested in pandoc.*

4 Submission Instructions

Submit via Canvas or to my email (grwells@uidaho.edu). Most assignments require a write-up and source code, both submitted in a compressed/zipped folder containing all files required to run. Write-ups must be submitted in PDF format. When submitting an assignment via email, please put the course number and the assignment name in the subject line. Submission requirements may be amended in assignment instructions.